PHASE 4 — PREDICTIVE & ADVANCED ANALYTICS

Real Estate AI Decision Intelligence Platform — Phase Completion Report

Prepared following the Master A–Z Development, Audit & Integration Prompt, Section 19 (Phase Completion Report) and Section 18 (Repository Audit Process).

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1. PHASE STATUS

PHASE 4 — COMPLETE. All seven candidate models (A–G) were built, trained/computed, evaluated against a documented baseline, and registered in the Model Registry with an honest, evidence-based status. Three of seven models are **not** approved for unqualified production use (two REJECTED, one CONDITIONAL, one UNCALIBRATED) — this is the correct, expected outcome of rigorous governance applied to this dataset, not a shortfall in the work.

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2. OBJECTIVE

Build validated predictive models that provide genuine business value (Master Prompt §4.1), obeying the Model Promotion Rule (§4.5): a model is not production-approved simply because it runs — it must demonstrate valid methodology, no leakage, acceptable performance, a useful baseline comparison, and real business value. Where it does not, the model is REJECTED or CONDITIONAL rather than silently accepted.

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3. REPOSITORY AUDIT SUMMARY (Section 18)

Before any Phase 4 code was written, the three delivered zip files were extracted and inspected:

Phase	Deliverable	Key findings used
P1	`phase1-realestates__clean_python_setupsql.zip`	14-table PostgreSQI (`sql/01_schema.sql`), raw→processed clea (`src/clean_real_esta` and a `docs/data_dict` Processed CSVs (`data/processed/*.cs` clean, ready-to-mod — used directly by F than raw. 20 SQL analytical vi (`analytics.v_localit` `v_property_comparabl` `v_rental_yield`, `v_demand_supply_tren`

P2	`real-estate-phase-2_sqlanalysis.zip`	confirms which loca KPIs are already reu side, so Phase 4 doe derive them, only co same underlying tab
P3	`real_estate_phase3-python-analytics.zip`	Full EDA/statistics r (`docs/eda_report.md`) **single most impor Phase 4.** It indepen flagged: (a) a **data-g bug** causing runawa compounding in `transactions.transac` `market_monthly.avera` (Section 9/12 of that that `days_on_market` **recoverable signal** anticipated as `REJEC` `area_sqft` dominates (r≈0.83–0.84), and (`property_type` is not significant price driv Wallis p=0.097).

No source data or Phase 1–3 code was modified. Phase 4 reads only from `data/processed/*.csv` (the Phase 1 cleaned output), consistent with Section 17 (Change Management: unrelated phases remain untouched).

Important scope note: the original synthetic-data generator script (`generate_data.py` / `generate_data_part2.py`, referenced in the Phase 1 README) was **not included** in any of the three delivered zips. This matters for Model C (see §6.3) — the Master Prompt's preferred remedy for the compounding bug ("fix generator → regenerate → rerun EDA → retrain") was not executable in this phase; the documented fallback (§4.6: "use a justified transformation/robust treatment") was applied instead.

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4. INPUTS

- `data/processed/properties.csv`, `projects.csv`, `localities.csv`, `cities.csv`, `developers.csv` (8,000 / 500 / 90 / 6 / 40 rows)
- `data/processed/transactions.csv`, `rentals.csv`, `listing_history.csv` (20,000 / 10,000 / 12,000 rows)
- `data/processed/market_monthly.csv` (6,480 rows = 90 localities × 72 months), `economic_monthly.csv` (72 months)
- Phase 3 EDA report and findings (`docs/eda_report.md`, `docs/eda_results_csv/*.csv`)

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5. WHAT WAS BUILT

A self-contained `phase4/` package:

```
phase4/
├── data/                        # copied from Phase 1
data/processed/*.csv (read-only input)
├── src/
│   ├── data_loader.py         # single source of truth: joins +
feature engineering
```

```
│   ├── governance.py          # ModelRegistry, metric bundles,
status rules
│   ├── model_a_valuation.py
│   ├── model_b_rent.py
│   ├── model_c_price_forecast.py
│   ├── model_d_demand_forecast.py
│   ├── model_e_dom.py
│   ├── model_f_sale_probability.py
│   ├── model_g_risk_score.py
│   ├── train_all.py           # orchestrator, idempotent re-run
│   └── test_phase4.py         # 14 automated tests
├── models/
│   ├── *.joblib                # trained model artifacts
│   ├── risk_score_v1_output.csv
│   └── model_registry.json     # machine-readable Model Registry
(Section 4.4 schema)
├── docs/
│   ├── phase4_report.md        (this file)
│   ├── model_registry.md
│   └── limitations.md
├── requirements-phase4.txt
└── README_PHASE4.md
```

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6. MODEL-BY-MODEL RESULTS

6.1 Model A — Property Valuation (`valuation_v1_gbr`)

Target: `properties.asking_price` · **Status: PASS WITH LIMITATIONS**

	R^2	MAE	RMSE	MA
GradientBoostingRegressor	**0.7955**	₹1,286,432	₹2,072,834	10.1
Linear Regression baseline	0.7983	₹1,297,460	₹2,058,447	10.5
Naive (median) baseline	-0.008	₹3,481,239	₹4,601,169	32.2

Both fitted models comfortably beat the naive baseline, but **the linear baseline is essentially tied with (marginally ahead of) the gradient-boosted model.** This is an honest finding, not a bug: Phase 3 EDA already showed `area_sqft` alone carries ~84% of feature importance and the price relationship is close to linear (r=0.838) — so a non-linear model earns little extra here. 59.2% of predictions land within 10% of actual price, just under the 60% "PASS" bar set for this model, hence **PASS WITH LIMITATIONS** rather than PASS. `monthly_rent` was deliberately excluded as a feature (cross-target leakage guard — see §7).

6.2 Model B — Rent Prediction (`rent_v1_gbr`)

Target: `properties.monthly_rent` (excluding 2% generator-imputed rows) · **Status: PASS WITH LIMITATIONS**

	R^2	MAE	RMSE	MAPE
GradientBoostingRegressor	0.6513	₹6,011	₹7,996	18.76%
Linear Regression baseline	0.6560	₹6,019	₹7,942	18.84%
Naive (median) baseline	-0.019	₹10,630	₹13,667	36.96%

Same pattern as Model A (area-dominated, near-linear relationship). Rent is inherently noisier than price in this dataset (lower R^2 ceiling) — documented, not hidden.

6.3 Model C — Price Forecasting, 6-month horizon (`price_forecast_v1_gbr_6m`)

Target: locality `average_price_sqft`, 6 months ahead, log1p-transformed · **Status: CONDITIONAL (forced)**

(log space)	R^2	MAE	RMSE	MAPE
GradientBoostingRegressor	0.9834	0.04	0.11	0.31%
Persistence baseline	0.9738	0.09	0.14	0.87%

The model beats a persistence (naive "no change") baseline on an honest **out-of-time split** (trained on months before 2024-07, tested on 2024-07 onward). **However, this model's status is forced to CONDITIONAL regardless of these strong metrics**, because Phase 3 EDA traced a real bug in the underlying `average_price_sqft` series: uncapped monthly compounding causes runaway growth in some localities (one locality: ₹13,592/sqft → ₹1,923,519/sqft, 2020→2025). The original data generator was not available to Phase 4 to fix at source, so the documented fallback (log-space modeling + honest out-of-time evaluation) was applied per Master Prompt §4.6. This model must be surfaced downstream as a **directional momentum signal only, never a precise price target.**

6.4 Model D — Demand Forecasting, 6-month horizon (`demand_forecast_v1_gbr_6m`)

Target: locality `demand_index`, 6 months ahead · **Status: CONDITIONAL**

	R^2	MAE	RMSE	MAPE
GradientBoostingRegressor	0.9320	2.96	3.73	3.79%
Persistence baseline	0.9330	2.88	3.70	3.70%

R^2 is high in absolute terms, but the model does **not** beat the persistence baseline (demand is highly autocorrelated month-to-month, so "no change" is already a very strong forecast). Per the promotion rule (Section 4.5: promotion requires a "useful baseline comparison"), failing to beat baseline downgrades this to **CONDITIONAL** rather than PASS WITH LIMITATIONS, despite the high R^2. A model that cannot outperform "assume nothing changes" is not yet demonstrating business value and needs human review before Phase 9's Prediction Agent may consume it.

6.5 Model E — Days-on-Market (`dom_v1_gbr`)

Target: `listing_history.days_on_market` (closed listings only) · **Status: REJECTED**

	R^2	MAE	RMSE
GradientBoostingRegressor	**-0.0096**	35.4 days	43.0 days
Naive (median) baseline	-0.0005	35.3 days	42.9 days

This independently **confirms** the Phase 3 EDA finding (which had already flagged DOM as `NEEDS_INVESTIGATION` / weak-signal). The model explains essentially zero variance beyond the mean and does not beat the naive baseline. Per §4.2 ("Only promote if useful predictive signal exists. Otherwise: REJECTED"), this model is REJECTED and must not be consumed by any downstream agent.

6.6 Model F — Sale Probability (`sale_probability_v1_gbr`)

Target: listing outcome, Sold vs Expired/Withdrawn (closed listings only) · **Status: REJECTED**

	ROC-AUC	PR-AUC	Brier	Calibration gap
GradientBoostingClassifier	**0.5197**	0.6198	0.2404	0.044
Logistic Regression baseline	0.508	0.626	0.238	—
Constant base-rate baseline	0.500	0.612	0.237	—

ROC-AUC of 0.52 is barely above chance (0.50) and below the 0.60 promotion floor. This is consistent with the same dataset pattern the Phase 3 EDA found elsewhere (e.g., `buyer_type` vs `transaction_type` independence, Chi-square p=0.94) — outcome/status fields in this synthetic dataset appear close to independent of the structural features available. REJECTED.

6.7 Model G — Risk Score, transparent composite (`risk_score_v1_composite`)

Not a trained model — deterministic weighted composite · **Status: UNCALIBRATED**

5 components, each 0–100 normalized: `valuation_uncertainty_risk` (25%, from Model A's residual gap), `developer_risk` (20%, inverse of rating/delivery/quality/track record), `locality_quality_risk` (20%, inverse of safety/infrastructure/development/connectivity), `market_liquidity_risk` (20%, from latest absorption/DOM/demand-supply gap), `price_volatility_risk` (15%, 12-month rolling volatility of locality price growth).

Weights, normalization method, and thresholds are fully documented in the registry (Section 4.2 requirement for transparent composites). Deliberately **excludes** Models E and F (both REJECTED) as inputs, per the Phase 9 agent-contract rule that rejected-model outputs must not be laundered into another score. Result across all 8,000 properties: mean risk score 35.6/100, band distribution LOW 27.2% / MODERATE 71.0% / ELEVATED 1.9% / HIGH 0.0%. Status is UNCALIBRATED because no real-world outcome label exists to validate against — the score is a **relative ranking tool**, not a calibrated probability.

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7. ML GOVERNANCE APPLIED (Section 4.3)

- **Leakage checks:** automated (`test_phase4.py`) — Model A asserts `monthly_rent` is never a feature; Model B asserts `asking_price` is never a feature; lag features in the locality-month panel are asserted to equal exactly the prior month's value (no future leakage in joins).
- **Temporal split:** used for Models C and D (out-of-time, cutoff 2024-07-01) since these are genuine forecasting tasks. Models A, B, E, F use random holdout since their targets are point-in-time attributes, not time series — this choice is stated explicitly in each registry entry rather than silently defaulted.
- **Baseline comparison:** every model is compared to a naive/persistence/constant baseline; Model D's failure to beat its baseline directly determined its status.
- **Censoring handled correctly:** Models E and F train only on *closed* listings (Sold/Expired/Withdrawn); `Active` listings are

excluded as censored (outcome not yet known), not silently treated as a class.
- **Imputed-target exclusion:** Model B excludes the 2% of `monthly_rent` rows that Phase 1's cleaning step median-imputed, so the model is not trained to reproduce an imputation rule.
- **Calibration:** measured for Model F (10-bin quantile calibration curve; gap=0.044) even though the model was ultimately rejected on discrimination (AUC) grounds.
- **Model Registry:** every model, including rejected ones, is registered with target, features, training data, validation method, metrics, baseline, status, limitations, business use, known failure cases, and version (Section 4.4 schema) — nothing is deleted or hidden.

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8. TESTS EXECUTED / PASSED / FAILED

`src/test_phase4.py` — **14/14 tests passed, 0 failed.**

Covers: no-duplicate-key joins, row-count integrity, lag-feature leakage guard, censored-listing exclusion, cross-target feature-leakage guards (Models A & B), artifact existence, prediction sanity (positive prices), risk-score bounds, registry JSON validity, registry status-value validity, and three "governance-outcome" tests that assert the REJECTED / CONDITIONAL / UNCALIBRATED statuses were actually applied (guards against a future re-run silently upgrading a model's status without a documented reason).

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9. VALIDATION RESULTS SUMMARY

Model	Status	Headline metric	Beats baseline?
A — Valuation	PASS WITH LIMITATIONS	R^2=0.7955	Yes (vs naive); ~tied with linear
B — Rent	PASS WITH LIMITATIONS	R^2=0.6513	Yes (vs naive); ~tied with linear
C — Price Forecast 6m	CONDITIONAL (forced)	R^2=0.9834 (log space)	Yes (vs persistence)
D — Demand Forecast 6m	CONDITIONAL	R^2=0.9320	**No** (vs persistence)
E — Days on Market	**REJECTED**	R^2=-0.0096	No
F — Sale Probability	**REJECTED**	ROC-AUC=0.52	Barely (vs 0.50)
G — Risk Score	UNCALIBRATED	n/a (composite)	n/a

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10. BUSINESS VALUE

- A defensible, baseline-checked valuation and rent estimate is now available to feed Phase 6's financial engine when a user has not supplied their own price/rent assumptions.
- A momentum signal (Model C/D) is available for the Decision Engine's market component — correctly labeled so it is never mistaken for a precise forecast.

- A transparent, auditable risk ranking (Model G) is available for Phase 7/9, built only from approved (non-rejected) inputs.
- Equally valuable: **two models were correctly rejected.** Preventing a near-random days-on-market or sale-probability prediction from reaching a user (and being mistaken for real signal) is itself a governance deliverable, directly satisfying Section 3.2 ("No Silent Guessing") and Section 4.5 (promotion rule).

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11. KNOWN LIMITATIONS (see also `docs/limitations.md`)

1. All models are trained on **synthetic data** — structurally realistic (per Phase 1 design) but not calibrated to any real market. No metric in this report should be read as a real-world accuracy claim.
2. Model C's target series carries a known, traced data-generation bug (uncapped compounding). The original generator was not available to Phase 4 to fix at source; a documented log-space fallback was used instead, and the model is CONDITIONAL by design.
3. Models E and F confirm — independently, via held-out evaluation — that `days_on_market` and sale outcome carry no learnable signal from the features available in this dataset. This is a data-generation-design limitation, not a modeling failure.
4. Model G (risk score) is UNCALIBRATED — a ranking tool, not a probability, until real historical outcomes exist to validate against.
5. No PostgreSQL connection was required or used in this phase; all Phase 4 work reads from Phase 1's `data/processed/*.csv` directly, since a live database instance was not part of the delivered environment. If a live `real_estate_db` is available, `src/data_loader.py` can be pointed at it in place of the CSVs with no change to any model script (the join logic is identical either way).

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12. DEPENDENCIES

- Upstream: Phase 1 (`data/processed/*.csv`, `docs/data_dictionary.md`), Phase 3 (`docs/eda_report.md` — directly informed Models C, E, F's governance decisions).
- Downstream: Phase 6 (Model A/B outputs as financial-engine inputs), Phase 7 (Model C/D momentum signals as risk drivers, Model G as a candidate risk-driver), Phase 9 (all `READY`/approved model outputs consumed by the Prediction Agent, which must respect the statuses recorded here — REJECTED models must never be silently used).

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13. SECURITY CONSIDERATIONS

No credentials are used or required (CSV-based data access in this phase). If/when Phase 4 is pointed at a live PostgreSQL instance, it must reuse Phase 1's `.env`-based credential pattern (Section 1.4) — no new credential-handling code was introduced.

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14. PERFORMANCE CONSIDERATIONS

Full `train_all.py` run (all 7 models, ~8,000 properties / 6,480 locality-months / 12,000 listings) completes in well under a minute on a single CPU core — no performance optimization was necessary at this data volume. Re-runs are idempotent (registry entries and joblib artifacts are overwritten per model_name).

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15. PRODUCTION READINESS

Per Master Prompt Section 24: this phase should be described as a **Production-Oriented Prototype**, not "Production Ready." Architecture, governance, and testing are in place; live-database integration, model monitoring/drift detection in production, CI/CD, and authentication/authorization for the eventual agent layer are Phase 9/10 and operational-hardening concerns, not yet built.

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16. NEXT PHASE DEPENDENCIES

Phase 5 (Evidence Intelligence & RAG) does not depend on Phase 4 directly (per the architecture diagram, P4 → P5 is sequential but P5's RAG layer is document-based, not model-based). Phases 6/7/8 (which may proceed in parallel once P5 is stable) will consume Model A/B/C/D/G outputs as described in §12. The Phase 9 Prediction Agent must load `models/model_registry.json` and refuse to serve REJECTED models' predictions.

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17. FINAL STATUS

PHASE 4: COMPLETE. 7/7 candidate models built and governed; 14/14 automated tests passing; Model Registry populated with full Section 4.4 schema for every model. Ready for Phase 5+ integration, with all limitations explicitly documented rather than hidden.